

2019 年第二学期《高等数学》考试试题参考解答

一、填空题

1、【答案】: $\left(-\frac{1}{7}, \frac{11}{7}, -\frac{5}{7}\right)$

【参考解析】: 求过点 $P(1, 1, 1)$ 与平面垂直的直线与平面的交点, 则与交点对称的点即为点关于平面对称的点. 直线方程为

$$\frac{x-1}{2} = \frac{y-1}{-1} = \frac{z-1}{3} = t$$

将 $x = 1 + 2t, y = 1 - t, z = 1 + 3t$ 代入平面方程, 得 $t = -\frac{2}{7}$, 即平面上的交点坐标为

$x = \frac{3}{7}, y = \frac{9}{7}, z = \frac{1}{7}$. 令对称点坐标为 (x_1, y_1, z_1) , 则

$$\frac{x_1+1}{2} = \frac{3}{7}, \frac{y_1+1}{2} = \frac{9}{7}, \frac{z_1+1}{2} = \frac{1}{7}$$

解得 $x_1 = -\frac{1}{7}, y_1 = \frac{11}{7}, z_1 = -\frac{5}{7}$.

2、【答案】: $-\frac{1}{2}$

【参考解析】: 直接对函数求两个二阶导数, 得

$$\begin{aligned}\frac{\partial z}{\partial x} &= \frac{y^2}{(x+y)^2}, \frac{\partial z}{\partial y} = \frac{x^2}{(x+y)^2} \\ \frac{\partial^2 z}{\partial x^2} &= -\frac{2y^2}{(x+y)^3}, \frac{\partial^2 z}{\partial y^2} = -\frac{2x^2}{(x+y)^3}\end{aligned}$$

代入所求式子和点的坐标, 得

$$\left[\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} \right]_{(1,1)} = \left[-\frac{2y^2 + 2x^2}{(x+y)^3} \right]_{(1,1)} = -\frac{2+2}{8} = -\frac{1}{2}$$

3、【答案】: $2x + z - 2 = 0$

【参考解析】: 令 $F(x, y, z) = x^2 + yz + z^2 - 1$, 则切平面的法向量为

$$\vec{n}(1, 1, 0) = (F'_x, F'_y, F'_z)_{(1,1,0)} = (2x, z, y + 2z)_{(1,1,0)} = (2, 0, 1)$$

所以切平面方程为

$$2(x-1) + 0(y-1) + 1(z-0) = 0, \text{ 即 } 2x - 2 + z = 0.$$

4、【答案】: $\frac{a}{2}$

【参考解析】: 由于被积函数定义在积分曲线上, 并且曲线关于 x 轴对称, 所以

RForest

仅供备考学习 严禁商用!

$$\oint_C \left(\frac{x^2}{2} + xy + y^2 \right) ds = \oint_C xy ds + \frac{1}{2} \oint_C (x^2 + 2y^2) ds$$

$$= 0 + \frac{1}{2} \oint_C ds = \frac{a}{2}$$

5、【答案】: $\frac{1}{2}$

【参考解析】: 考虑函数为周期 $T = 4$ 的傅里叶级数, 并且函数为偶函数, 所以

$$S(2019) = S(505 \times 4 - 1) = S(-1) = S(1)$$

$$= \frac{f(1+) + f(1-)}{2} = \frac{1+0}{2} = \frac{1}{2}$$

二、选择题

6、【答案】: 【C】

【参考解析】: 由点坐标所对应的参数值为 $t = -1$, 曲线的切向量为

$$T(-1) = (x'(-1), y'(-1), z'(-1)) = (1, -2, 3)$$

所以切线方程为 $\frac{x}{1} = \frac{y-2}{-2} = \frac{z}{3}$, 即正确选项为【C】.

7、【答案】: 【D】

【参考解析】: 交换积分次序得

$$I = \int_0^1 dy \int_0^y f(x)f(y) dx = \int_0^1 dx \int_0^x f(x)f(y) dy$$

所以由积分符号的无关性和积分的线性运算性质, 得

$$I = \frac{1}{2} \left[\int_0^1 dx \int_x^1 f(x)f(y) dy + \int_0^1 dx \int_0^x f(x)f(y) dy \right]$$

$$= \frac{1}{2} \int_0^1 dx \int_0^1 f(x)f(y) dy = \frac{1}{2} \int_0^1 f(x) dx \int_0^1 f(y) dy = \frac{1}{2} \left[\int_0^1 f(x) dx \right]^2$$

即正确选项为【D】.

8、【答案】: 【B】

【参考解析】: 从图可以看出, 沿 x 轴正方向移动, 函数值增加, 沿 y 轴正方向函数值减小, 所以 $f'_x(x_0, y_0) > 0, f'_y(x_0, y_0) < 0$, 即正确选项为【B】.

9、【答案】: 【A】

【参考解析】: 微分方程对应的特征方程为 $r^2 + br + 1 = 0$, 解得

$$r_{1,2} = \frac{-b \pm \sqrt{b^2 - 4}}{2}$$

当 $|b| \geq 2$, 则 $b > \sqrt{b^2 - 4}$, 则当 $b > 2$ 时 $r_{1,2} = \frac{-b \pm \sqrt{b^2 - 4}}{2} < 0$, 则

$$y = C_1 e^{r_1 x} + C_2 e^{r_2 x} \rightarrow 0 (x \rightarrow +\infty)$$

RForest

仅供备考学习 严禁商用!

所以有界. 当 $b < 2$ 时 $r_{1,2} = \frac{-b \pm \sqrt{b^2 - 4}}{2} > 0$, 则

$$y = C_1 e^{r_1 x} + C_2 e^{r_2 x} \rightarrow \infty (x \rightarrow +\infty)$$

无界. 当 $b = 2$ 时, $r_{1,2} = -1 < 0$, 则

$$y = C_1 e^{-x} + C_2 x e^{-x} \rightarrow 0 (x \rightarrow +\infty)$$

有界. 当 $b = -2$ 时, $r_{1,2} = 1 < 0$, 则

$$y = C_1 e^x + C_2 x e^x \rightarrow \infty (x \rightarrow +\infty)$$

无界. 所以 (B) (C) (D) 错误! 正确选项为 【A】.

10、【答案】: 【A】

【参考解析】: $R = \lim_{n \rightarrow \infty} \left| \frac{a_n}{a_{n+1}} \right|$, 收敛区间为 $(1-R, 1+R)$, 则 $R = 2$. 又由于

$$\lim_{n \rightarrow \infty} \left| \frac{\frac{a_n}{2^n}}{\frac{a_{n+1}}{2^{n+1}}} \right| = \lim_{n \rightarrow \infty} \left| \frac{a_n}{2^n} \cdot \frac{2^{n+1}}{a_{n+1}} \right| = 2 \lim_{n \rightarrow \infty} \left| \frac{a_n}{a_{n+1}} \right| = 2R = 4$$

所以收敛区间为 $(-5, 3)$, 因此 $x = -3$ 处绝对收敛, 即正确选项为 【A】.

三、解答与证明题

11、【参考解析】: 记 $F(x, y, z) = f(yz, y - x)$, 则

$$\begin{aligned} F'_x &= f'_1 \cdot 0 + f'_2 \cdot (-1) = -f'_2 \\ F'_y &= f'_1 \cdot z + f'_2 \cdot 1 = z f'_1 + f'_2 \\ F'_z &= f'_1 \cdot y + f'_2 \cdot 0 = y f'_1 \end{aligned}$$

所以得 $\frac{\partial z}{\partial x} = -\frac{F'_x}{F'_z} = \frac{f'_2}{y f'_1}$, $\frac{\partial z}{\partial y} = -\frac{F'_y}{F'_z} = -\frac{z f'_1 + f'_2}{y f'_1}$.

12、【参考解析】:

$$\begin{aligned} \operatorname{div}(u \vec{A}) &= \frac{\partial(uP)}{\partial x} + \frac{\partial(uQ)}{\partial y} + \frac{\partial(uR)}{\partial z} \\ &= P \frac{\partial u}{\partial x} + u \frac{\partial P}{\partial x} + Q \frac{\partial u}{\partial y} + u \frac{\partial Q}{\partial y} + R \frac{\partial u}{\partial z} + u \frac{\partial R}{\partial z} \\ &= u \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z} \right) + \left(\frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}, \frac{\partial u}{\partial z} \right) \cdot (P, Q, R) \\ &= u \operatorname{div} \vec{A} + \operatorname{grad} u \cdot \vec{A} \end{aligned}$$

13、【参考解析】: 根据 Ω 的对称性, 得

$$\iiint_{\Omega} \frac{x}{1+x^2+y^2+z^2} dV = \iiint_{\Omega} \frac{y}{1+x^2+y^2+z^2} dV = 0$$

由三重积分的球坐标计算方法, 令 $x = r \sin \varphi \cos \theta$, $y = r \sin \varphi \sin \theta$, $z = r \cos \varphi$, 则

仅供备考学习 严禁商用!

$$\Omega: \theta \in [0, 2\pi], \varphi \in \left[0, \frac{\pi}{2}\right], 0 \leq r \leq 1$$

因此

$$\begin{aligned} I &= \iiint_{\Omega} \frac{z}{1+x^2+y^2+z^2} dV = \int_0^{2\pi} d\theta \int_0^{\frac{\pi}{2}} d\varphi \int_0^1 \frac{r \cos \varphi}{1+r^2} \cdot r^2 \sin \varphi dr \\ &= \int_0^{2\pi} d\theta \int_0^{\frac{\pi}{2}} (\sin \varphi \cos \varphi) d\varphi \int_0^1 \frac{r^3}{1+r^2} dr = 2\pi \cdot \frac{1}{2} \int_0^1 \left(r - \frac{r}{1+r^2}\right) dr \\ &= 2\pi \cdot \frac{1}{2} \cdot \frac{1}{2} (1 - \ln 2) = \frac{\pi}{2} (1 - \ln 2) \end{aligned}$$

14、【参考解析】：由格林公式可知

$$\begin{aligned} \oint_C (x - y \sin x) dx + (xy + \cos x) dy &= \iint_D y dx dy \\ &= \int_0^{\pi} dx \int_0^{\sin x} y dy = \frac{1}{2} \int_0^{\pi} \sin^2 x dx = \frac{\pi}{4} \end{aligned}$$

15、【参考解析】：(1) 因为

$$\frac{\partial f}{\partial x} = \frac{2x}{x^2 + y^2 + z^2}, \frac{\partial f}{\partial y} = \frac{2y}{x^2 + y^2 + z^2}, \frac{\partial f}{\partial z} = \frac{2z}{x^2 + y^2 + z^2}$$

所以函数的全微分为

$$df = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy + \frac{\partial f}{\partial z} dz = 2 \frac{x dx + y dy + z dz}{x^2 + y^2 + z^2}$$

(2) 由于 $\left. \frac{\partial f}{\partial x} \right|_{(1,1,1)} = \frac{2}{3}$, $\left. \frac{\partial f}{\partial y} \right|_{(1,1,1)} = \frac{2}{3}$, $\left. \frac{\partial f}{\partial z} \right|_{(1,1,1)} = \frac{2}{3}$, 又 $\vec{u} = (3, 0, 4)$, 则

$$\vec{u}^\circ = (\cos \alpha, \cos \beta, \cos \gamma) = \left(\frac{3}{5}, 0, \frac{4}{5}\right)$$

于是可得方向导数为

$$\begin{aligned} D_{\vec{u}} f|_{(1,1,1)} &= \left. \frac{\partial f}{\partial x} \right|_{(1,1,1)} \cos \alpha + \left. \frac{\partial f}{\partial y} \right|_{(1,1,1)} \cos \beta + \left. \frac{\partial f}{\partial z} \right|_{(1,1,1)} \cos \gamma \\ &= \frac{2}{3} \cdot \frac{3}{5} + \frac{2}{3} \cdot 0 + \frac{2}{3} \cdot \frac{4}{5} = \frac{14}{15} \end{aligned}$$

16、【参考解析】：令 C_1 为点 $(1, -1)$ 沿抛物线 $x = y^2$ 到原点的曲线, C_2 是从原点出发沿 y 轴到点 $(0, 1)$ 的曲线, 则

$$\begin{aligned} W &= \int_{C_1+C_2} (x + y^2) dx + (x - y) dy \\ &= \int_{C_1} (x + y^2) dx + (x - y) dy + \int_{C_2} (x + y^2) dx + (x - y) dy \end{aligned}$$

由对坐标的曲线积分的直接算法, 得

RForest

仅供备考学习 严禁商用!

$$\int_{C_1} (x+y^2) dx + (x-y) dy = \int_{-1}^0 (4y^3 + y^2 - y) dy = -\frac{1}{6}$$

$$\int_{C_2} (x+y^2) dx + (x-y) dy = \int_0^1 (-y) dy = -\frac{1}{2}$$

$$\text{所以 } W = \int_{C_1+C_2} (x+y^2) dx + (x-y) dy = -\frac{1}{6} - \frac{1}{2} = -\frac{2}{3}.$$

17、【参考解析】: (1) 由于微分式为全微分方程, 所以

$$\frac{\partial}{\partial y} ([e^{-x} - f(x)]y) = \frac{\partial}{\partial x} (f(x))$$

即 $e^{-x} - f(x) = f'(x)$. 问题归结为求初值问题

$$\begin{cases} f'(x) + f(x) = e^{-x} \\ f(0) = 1 \end{cases}$$

解得 $f(x) = (x+1)e^{-x}$.

(2) 由 $e^{-x} - f(x) = f'(x)$ 得

$$[e^{-x} - f(x)]y dx + f(x) dy = f'(x)y dx + f(x) dy = d(yf(x))$$

因此全文方程的通解为 $yf(x) = C$, 即 $y(x+1)e^{-x} = C$.

18、【参考解析】: 令 $u_n(x) = \frac{x^{2n-1}}{2n-1}$, 则由比值判别法, 得

$$\lim_{n \rightarrow +\infty} \left| \frac{u_{n+1}(x)}{u_n(x)} \right| = \lim_{n \rightarrow +\infty} \left| \frac{x^{2n+1}}{2n+1} \frac{2n-1}{x^{2n-1}} \right| = x^2$$

令 $x^2 < 1$, 得收敛区间 $(-1, 1)$. 当 $x = \pm 1$, 幂级数为 $\pm \sum_{n=1}^{\infty} \frac{1}{2n-1}$ 发散, 所以收敛域为 $(-1, 1)$.

记幂级数的和函数为 $S(x)$, 则逐项求导得

$$S'(x) = 1 + x^2 + x^4 + \cdots + x^{2n-2} + x^{2n} + \cdots = \frac{1}{1-x^2}, \quad -1 < x < 1$$

注意到 $S(0) = 0$, 于是再两端积分, 得

$$S(x) = \frac{1}{2} \ln \left(\frac{1+x}{1-x} \right), \quad -1 < x < 1$$

极爱那个 $x = \frac{1}{3}$ 代入, 则得

$$\sum_{n=1}^{\infty} \frac{1}{(2n-1) \cdot 3^{2n-1}} = \frac{1}{2} \ln \left(\frac{1+\frac{1}{3}}{1-\frac{1}{3}} \right) = \frac{1}{2} \ln 2$$

19、【参考解析】: (1) 由散度计算公式, 得

RForest

仅供备考学习 严禁商用!

$$\begin{aligned}\operatorname{div} \vec{v} &= \frac{\partial}{\partial x} \left[-\frac{y+xz}{(z^2+1)^2} \right] + \frac{\partial}{\partial y} \left[-\frac{yz-x}{(z^2+1)^2} \right] + \frac{\partial}{\partial z} \left[-\frac{1}{z^2+1} - x^2 \right] \\ &= -\frac{z}{(z^2+1)^2} - \frac{z}{(z^2+1)^2} + \frac{2z}{(z^2+1)^2} = 0\end{aligned}$$

(2) 所求流量为

$$\Phi = \iint_S -\frac{y+xz}{(z^2+1)^2} dy dz - \frac{yz-x}{(z^2+1)^2} dz dx - \left(x^2 + \frac{1}{z^2+1} \right) dx dy$$

添加辅助面 $S_0: z=0, (x,y) \in D: x^2+y^2 \leq 1$, 法向量指向上侧, 则 S_0+S 构成闭曲面, 设其围成的区域为 Ω , 则由高斯公式, 得

$$\oint_{S+S_0} = \iiint_{\Omega} \operatorname{div} \vec{v} dV = 0$$

于是可得

$$\begin{aligned}\Phi &= \iint_S = -\iint_{S_0} \\ &= \iint_{S_0} \left(x^2 + \frac{1}{z^2+1} \right) dx dy = \iint_D (1+x^2) dx dy \\ &= \iint_D dx dy + \frac{1}{2} \iint_D (x^2+y^2) dx dy = \pi + \frac{1}{2} \int_0^{2\pi} d\theta \int_0^1 \rho^2 \cdot \rho d\rho \\ &= \pi + \frac{1}{2} \cdot 2\pi \cdot \frac{1}{4} = \pi + \frac{\pi}{4} = \frac{5\pi}{4} \quad (\text{cm}^3/s)\end{aligned}$$

20、【参考解析】: 由拉格朗日乘数法, 令

$$L(x, y, z, \lambda) = xyz^3 + \lambda(x+y+z-1)$$

解方程组

$$\begin{cases} L'_x = yz^3 + \lambda = 0 \\ L'_y = xz^3 + \lambda = 0 \\ L'_z = 3xyz^2 + \lambda = 0 \\ L'_\lambda = x+y+z-1 = 0 \end{cases}$$

由前面两个方程相减得 $(x-y)z^3 = 0$. 由于 $x > 0, y > 0, z > 0$. 因此 $x = y$. 同理, 由第一个方程和第三个方程相减得 $z = 3x$. 将两式代入第四个方程, 解得

$$x = \frac{1}{5}, \quad y = \frac{1}{5}, \quad z = \frac{3}{5}$$

由于函数 $f(x, y, z) = xyz^3$ 在平面 $x+y+z=1$ 位于第一卦限的部分上连续, 而在这部分的边界线, 即 $z=0, x+y=1$ 或 $y=0, x+z=1$ 或 $x=0, z+y=1$ 上, $f(x, y, z) = 0$. 所以函数

故函数 $f(x, y, z) = xyz^3$ 的最大值只能在这部分的内部达到, 又 $\left(\frac{1}{5}, \frac{1}{5}, \frac{3}{5}\right)$ 为唯一可疑点, 故函数

RForest

仅供备考学习 严禁商用!

$f(x, y, z) = xyz^3$ 在约束条件 $x + y + z = 1$ ($x > 0, y > 0, z > 0$) 的最大值为

$$f\left(\frac{1}{5}, \frac{1}{5}, \frac{3}{5}\right) = \frac{27}{5^5} = \frac{27}{3125}$$

由上面的结论可知, 当 $x + y + z = 1$ ($x > 0, y > 0, z > 0$) 时有

$$f(x, y, z) = xyz^3 \leq \frac{27}{5^5}$$

则对任意正数 a, b, c , 令

$$x_0 = \frac{a}{a+b+c}, y_0 = \frac{b}{a+b+c}, z_0 = \frac{c}{a+b+c},$$

则 $x_0 + y_0 + z_0 = 1$, 因此

$$f(x_0, y_0, z_0) = x_0 y_0 z_0^3 = \frac{abc^3}{(a+b+c)^5} \leq \frac{27}{5^5}, \text{ 即 } abc^3 \leq 27 \left(\frac{a+b+c}{5} \right)^5.$$

RForest

仅供备考学习 严禁商用！

第 7 页 (共 7 页)